

WEEK 8 – Dashboard Creation & Business Visualization

Objective

To transform quantitative risk metrics into interactive business intelligence dashboard.

1. Tableau Data Connection

Connected 3 CSV files:

- week5_finance_cleaned.csv
- rolling_volatility.csv
- var_results.csv

Ensured:

- Date field recognized as Date type
 - Percentage formatting applied correctly
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2. Dashboard Components Created

Portfolio Return Trend

- Line chart
 - Shows daily return movement
 - Helps identify profit/loss cycles
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Rolling Volatility Chart

- Line chart
 - Displays 30-day standard deviation
 - Highlights uncertainty spikes
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Value at Risk KPI

- Large percentage display
- Risk indicator
- Negative value formatted as %

Date Filter

- Added interactive date range filter
 - Ensured dashboard components update dynamically
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Interactivity Testing

Validated:

- Clicking date filter updates all charts
 - No lag in dashboard refresh
 - Proper cross-sheet behavior
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Visual Formatting

Applied:

- Percentage formatting (2 decimals)
 - Clean axis labels
 - Dashboard title:
 - AlphaPulse – Portfolio Risk & Return Dashboard
 - Professional layout alignment
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Business Interpretation Layer

Dashboard now enables:

- ✓ Risk Monitoring
- ✓ Downside Exposure Analysis
- ✓ Market Uncertainty Tracking
- ✓ Quick Decision Support

FINAL PROJECT STATUS

Week	Status
Week 5	Data Collection of 2 nd project and Quantitative Modeling
Week 6	Visualization
Week 7	Risk Validation & Final Output
Week 8	Dashboard Deployment

Conclusion:

In Week 7, I implemented the final financial risk modeling pipeline including portfolio return calculation, rolling volatility, Monte Carlo simulation, and Value at Risk. I generated structured CSV outputs for BI integration.

In Week 8, I built an interactive Tableau dashboard integrating these outputs, applied percentage formatting, added dynamic date filters, and designed KPI visualization for business-level decision making.